Drone wiring | Power

Every airborne electrical part, with the four motor power branches shown separately.

Battery + | Regulated +5 V | Regulated +3.3 V | GND / battery -

ManiaX 3S LiPo

850 mAh, XT60

11.1 V nominal

12.6 V fully charged

\+

\-

Smoke stopper

Bench use only

Remove for flight

Matek HCS-150A

Positive lead through the sensing hole once

Keep battery - outside the hole. Follow the sensor current-direction marking.

Matek XT60 PDB

Raw battery distribution

5 V and 12 V BEC pads:
NC in this design

BAT+

BAT- / GND

470 uF / 35 V

Across BAT+ and BAT-

Striped leg to BAT-

3 phases

ESC 1 EMAX 12 A 2-3S input	**M1** 2204
ESC 2 EMAX 12 A 2-3S input	**M2** 2204
ESC 3 EMAX 12 A 2-3S input	**M3** 2204
ESC 4 EMAX 12 A 2-3S input	**M4** 2204

One ESC per motor. Motor phase wires never connect to the STM32.

Waveshare 5 V / 4 A buck

VIN from PDB BAT+; GND from PDB BAT-

Leave configured for 5 V output.

VOUT = +5 V rail; GND = common GND

PDB BAT+

PDB BAT-

HCS-150A small pads

Vbat -> PDB BAT+ 5V -> +5 V G -> GND

Volt / Curr -> STM32 ADC inputs, see sheet 2.

Electronics power connections

Each listed GND returns to the same common ground.

+5 V FROM WAVESHARE VOUT

STM32 core board	Verified 5 V input only *
Foxeer M10Q 120 V2	5V / VCC
RadioMaster RP1 V2	5V
Matek HCS-150A	5V pad
VIFLY Finder Mini	Red / 5V lead

+3.3 V FROM VERIFIED CORE-BOARD REGULATOR *

SparkFun LSM6DSO	3V3
Waveshare BMP388	VCC
Pi Hut MicroSD breakout	3v pad; insert card in socket

* Confirm the core-board 5 V input, 3.3 V output and regulator current headroom. SD-card write bursts must not brown out the controller. If insufficient, add a suitable separate 3.3 V regulator.

Do not join the ESC BEC outputs or PDB BEC outputs to the Waveshare +5 V rail. Never feed raw battery voltage into the STM32 or sensors.

Drone wiring | Signals

Use these named functions when assigning STM32 pins in firmware. They are not verified physical header labels.

SparkFun LSM6DSO

3V3 -> +3.3 V; GND -> GND

SCL / SCK	IMU_SPI_SCK
SDA / SDI	IMU_SPI_MOSI
SDO	IMU_SPI_MISO
CS	IMU_CS
INT1	IMU_DRDY input

SPI: cut the default address jumper. INT2 and auxiliary pins unused.

Waveshare BMP388

VCC -> +3.3 V; GND -> GND

SCL	BARO_I2C_SCL
SDA	BARO_I2C_SDA

Use its default I2C mode. INT unused.

Pi Hut MicroSD breakout

3v -> +3.3 V; G -> GND

CLK	SD_SPI_SCK
DI / CMD	SD_SPI_MOSI
DO / DAT0	SD_SPI_MISO
CS / DAT3	SD_CS

Separate SPI bus. DT1, DT2, CD unused; retain onboard pull-ups.

EC Buying STM32H743

3.3 V logic

Exact GPIO / header map pending

GPS_UART_RX	GPS TX
GPS_UART_TX	GPS RX
MAG_I2C_SCL	SCL
MAG_I2C_SDA	SDA
RC_UART_RX	TX / T
RC_UART_TX	RX / R
ADC_CURRENT	Curr
ADC_VOLTAGE	Volt
BUZZER_GPIO	Yellow / BZ through driver on sheet 3

Foxeer GPS + IST8310

5V -> +5 V; GND -> GND

GPS TX
GPS RX
SCL
SDA

Use a separate I2C bus for compass. Verify cable pin order and logic level.

RadioMaster RP1 V2

5V -> +5 V; GND -> GND

TX / T
RX / R

CRSF on one full UART. Attach antenna. Pocket connects by radio only.

Matek HCS-150A

5V -> +5 V; G -> GND; Vbat -> BAT+

Curr
Volt

Analogue signals, 0-3.3 V. TX / RX pads unused.

VIFLY Finder Mini

Yellow / BZ through driver on sheet 3

MOTOR_1..4 -> matching ESC signal wires

ESC signal GND -> STM32 GND

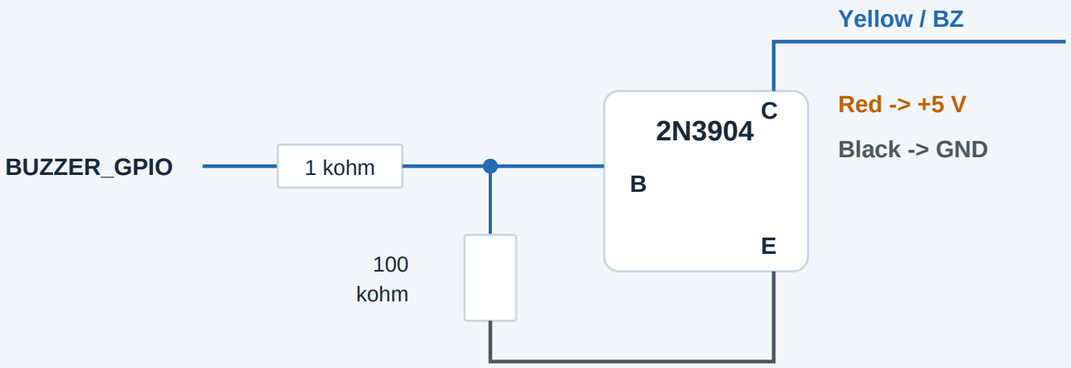
All four thin red ESC BEC wires -> NC

Drone wiring | Buzzer, bench equipment and checks

The transistor circuit is an added part. Ground equipment disconnects before flight.

Add a low-side buzzer driver

Example: 2N3904 NPN + 1 kohm + 100 kohm resistors



GPIO high pulls BZ low. Confirm the transistor pin order for its package.

Off-drone connections

ST-Link V2 compatible

SWDIO -> core SWDIO; SWCLK -> SWCLK; GND -> GND. NRST if available. A true VTref input senses target 3.3 V; a clone 3.3 V power output stays disconnected.

Pocket + two 18650 cells

Install cells in the transmitter with the marked polarity. Pocket ELRS 2.4 GHz links wirelessly to RP1; there is no cable to the drone.

HTRC B3 charger

With LiPo disconnected from drone: 3S balance lead -> charger 3S socket, per its manual. Main XT60 stays disconnected; do not use a 2S socket.

Resolve before soldering or applying power

STM32 board and firmware

Confirm the board schematic, its 5 V input and 3.3 V rail. Assign two SPI buses, two I2C buses, two UARTs, four timer outputs, two ADC inputs, IMU interrupt and buzzer GPIO. Reserve SWD pins. Stock flight firmware is not guaranteed to support this core board or these sensors.

ESC wiring and protocol

Each ESC: thick red -> PDB BAT+; thick black -> PDB BAT-; three phase wires -> its motor. Signal + signal GND -> controller. Remove and insulate every thin red 5 V BEC lead. Confirm 3.3 V signal compatibility. Use supported PWM; do not assume DShot.

Logic, rails and first power-up

Keep sensor / SD logic at 3.3 V. Check the Foxeer UART and I2C high levels; add level translation if required. Check I2C pull-ups and avoid redundant low-value pull-ups. With props off, check polarity, shorts, 5 V and 3.3 V, then power through the smoke stopper. Avoid USB / debugger backfeeding.

Motor order and mechanical parts

Map ESC 1-4 to the motor positions used by your firmware. Verify rotation with props removed; swap any two phase wires with power off if needed. Fit matching props only after setup. Frame, mounts, strap, fixings and spare props have no signal wiring. Keep the LiPo restrained and all cables clear of props.

The included 5045 three-blade props have no matched current test in the listing. Verify load current stays within each 12 A ESC rating. HCS-150A scale: I = Vcurr / 0.020; Vbattery = 21 x Vvolt. At a full 3S charge Vvolt is about 0.60 V. Calibrate against measurements.

Source documents and product references

- SparkFun LSM6DSO hookup guide
- Matek HCS-150A pin definitions
- Waveshare BMP388 documentation
- Pi Hut MicroSD breakout
- Waveshare buck documentation
- RadioMaster RP1 specifications
- ExpressLRS receiver wiring
- VIFLY Finder Mini manual
- Foxeer supplier specifications
- Exact motor / ESC set supplied
- STM32 listing, schematic not verified
- EMAX 12 A BLHeli product page